Rodrigo Roque-Hernandez

919-348-3083 | rodrigoroque456@gmail.com | linkedin.com/in/rroque-hernandez

Applied AI & Data Systems Builder — Human-Centered Product Design

Education

University of North Carolina at Chapel Hill — Chapel Hill, NC
Bachelor of Science in Information Science & Data Science — *August 2023 – May 2027*
Awards & Activities: — *HSF Scholarhip Award, SHPE, CAMpFIRE Scholarship*

Technical Experience

Director of Web Development / Technical Lead — Aug 2025 – Present
Multi-Org Platforms — *–React, Firebase, HTML, CSS, JavaScript, Figma*

- Led cross-functional teams of developers and designers, driving frontend, backend, and UX execution while coordinating technical direction across multi-organization platforms
- Designed UX workflows and modular interface systems supporting event registration, content publishing, and hackathon platforms
- Integrated analytics tracking and platform optimization strategies, increasing user engagement and session duration by 23%

ecDNA Research Assistant — Oct. 2024 – May 2025
Brunk Cytogenomics Lab — *–Python, PyTorch, OpenCV, Pandas*

- Developed deep learning computer vision pipelines to detect and localize extra-chromosomal DNA (ecDNA) in FISH microscopy images, improving analysis efficiency by 25%
- Built automated preprocessing workflows including normalization, augmentation, and coordinate mapping to scale biomedical imaging datasets
- Quantified model performance and structural patterns in cancer imaging data, reducing manual review time by 12%

Computational Research Assistant — Sep 2024 – May 2025
Baker Lab — *– Python, Unix, Jupyter Notebooks, Unipressed*

- Automated ingestion and processing of 200+ protein records (UniProt, PDB) to enhance AlphaFold-based structural analysis pipelines
- Built data visualization workflows supporting biological inference and experimental decision-making
- Contributed 50K+ lines of code to an established codebase via Git
- Optimized reproducible data workflows for large-scale biological datasets

Engineering & Product Projects

Diabetes Readmission Prediction Model | *Python, scikit-learn,*

- Built Decision Tree and Random Forest models to predict 30-day hospital readmission risk using structured clinical datasets
- Designed preprocessing pipelines using ColumnTransformer to handle missing values and categorical features
- Evaluated performance with ROC curves and classification metrics (Test AUC = 0.656)

SolHacks Platform | *Figma, React, UX Architecture*

- Designed full mobile and desktop interface system including onboarding flows, schedules, and team management modules
- Built reusable component structures bridging product design and frontend engineering workflows
- Applied usability-driven interaction design to support a 150+ attendee hackathon platform
- Collaborated with organizing leadership and engineering teams to align platform design with event strategy and participant experience goals

Technical Skills

Languages: Python, R, JavaScript, SQL, HTML/CSS
Frameworks: React, WordPress
Developer Tools: Git, VS Code, JupyterLab, Unix, Google Colab
Libraries: PyTorch, scikit-learn, pandas, NumPy, Matplotlib, seaborn
Design & Product: Figma, UX Prototyping, Component Systems